$\boxed{\text{Def}}$

$$v0 = \mathrm{var}\ vz$$
$$v1 = \mathrm{var}(vs\ vz)$$
$$v2 = \mathrm{var}(vs(vs\ vz))$$

$$\mathrm{var}\ x\ [p] = \mathrm{var}(vs\ x)$$

$$\underbrace{\mathrm{var}\ vz}_{:\ \mathrm{Tm}\ (\Gamma \triangleright A)\ A}\ [\langle t \rangle] = t$$

$: \mathrm{Tm}\ \Gamma\ A$

$: \mathrm{Sub}\ \Gamma\ (\Gamma \triangleright A)$

$: \mathrm{Tm}\ \Gamma\ A$

$$\mathrm{var}(vs\ x)\ [\langle t \rangle] = \mathrm{var}\ x$$

$x : \mathrm{Var}\ \Gamma\ A$

$vs\ x : \mathrm{Var}\ (\Gamma \triangleright B)\ A$

$\mathrm{var}(vs\ x) : \mathrm{Tm}\ (\Gamma \triangleright B)\ A$

$\langle t \rangle : \mathrm{Sub}\ \Gamma\ (\Gamma \triangleright B)$

$t : \mathrm{Tm}\ \Gamma\ B$

$\mathrm{var}(vs\ x)\ [\langle t \rangle] : \mathrm{Tm}\ \Gamma\ A$

$_[_] :$

$\mathrm{Tm}\ \Delta\ A \to$

$\mathrm{Sub}\ \Gamma\ \Delta \to$

$\mathrm{Tm}\ \Gamma\ A$

$p : \mathrm{Sub}\ (\Gamma \triangleright A)\ \Gamma$

$$\frac{t : \mathrm{Tm}\ \Gamma\ A}{\langle t \rangle : \mathrm{Sub}\ \Gamma\ (\Gamma \triangleright A)}$$

1 $\mathrm{var}\ x\,[p] = \mathrm{var}\,(\mathrm{vs}\ x)$

2 $\mathrm{var}\ \mathrm{vt}\,[\langle t\rangle] = t$

3 $\mathrm{var}\,(\mathrm{vs}\ x)\,[\langle t\rangle] = \mathrm{var}\ x$

4 $\mathrm{var}\ \mathrm{vt}\,[\sigma^{+}] = \mathrm{var}\ \mathrm{vt}$

5 $\mathrm{var}\,(\mathrm{vs}\ x)\,[\sigma^{+}] = \underline{\mathrm{var}\ x\,[\sigma]}\,[p]$

$$\mathrm{var}\,(\mathrm{vs}\ \mathrm{vt})\,[\langle t\rangle^{+}] \overset{5}{=} \mathrm{var}\ \mathrm{vt}\,[\langle t\rangle]\,[p] \overset{4}{=} t\,[p]$$

$$\mathrm{var}\ \mathrm{vt}\,[\langle t\rangle^{+}] \overset{4}{=} \mathrm{var}\ \mathrm{vt}$$

$$\underline{\mathrm{var}\,(\mathrm{vs}\ \mathrm{vt})\,[\langle t\rangle^{++}]} \overset{5}{=} \underline{\mathrm{var}\ \mathrm{vt}\,[\langle t\rangle^{+}]}\,[p] \overset{4}{=} \mathrm{var}\ \mathrm{vt}\,[p] \overset{1}{=} \mathrm{var}\,(\mathrm{vs}\ \mathrm{vt})$$

$$\mathrm{var}\,(\mathrm{vs}\,(\mathrm{vs}\ \mathrm{vt}))\,[p^{+}] \overset{5}{=}$$

$$\mathrm{var}\,(\mathrm{vs}\ \mathrm{vt})\,[p]\,[p] \overset{1,1}{=} \mathrm{var}\,(\mathrm{vs}\,(\mathrm{vs}\ \mathrm{vt}))$$

$$\mathrm{var}\,(\mathrm{vs}^{2}\ \mathrm{vt})\,[p^{++}] \overset{5}{=} \mathrm{var}\,(\mathrm{vs}\ \mathrm{vt})\,[p^{+}]\,[p] = \mathrm{var}\ \mathrm{vt}\,[p]\,[p]\,[p]$$

$$\mathrm{var}\,(\mathrm{vs}^{2}\ \mathrm{vt})\,[p^{+++}] =$$

$$\frac{\sigma : \mathrm{Sub}\ \Delta\ \Gamma}{\sigma^{+} : \mathrm{Sub}\,(\Delta \triangleright A)\,(\Gamma \triangleright A)}$$

$$p^{t.n+} \qquad \langle t\rangle^{t.n+}$$

$t : \mathrm{Tm}\ \Gamma\ A$

$$: \mathrm{Sub}\,(\Gamma \triangleright A \triangleright B_1 \triangleright \ldots \triangleright B_n)\,(\Gamma \triangleright B_1 \triangleright \ldots \triangleright B_n)$$

$$: \mathrm{Sub}\,(\Gamma \triangleright B_1 \triangleright \ldots \triangleright B_n)\,(\Gamma \triangleright A \triangleright B_1 \triangleright \ldots \triangleright B_n)$$

$$\mathrm{var}\ vt\,[\langle t\rangle] \overset{\sim}{\underset{2}{=}} t$$

$$\mathrm{var}\,(\mathrm{in}\ vt)\,[\langle t\rangle] \overset{3}{=} \mathrm{var}\ vt$$

$$\mathrm{var}\,(\mathrm{in}\,(\mathrm{in}\ vt))\,[\langle t\rangle] \overset{3}{=} \mathrm{var}\,(\mathrm{in}\ vt)$$

$$\mathrm{def}\ t\ u\,[G] =$$

$$\mathrm{def}\ (t\,[G])\,(u\,[G\,t])$$

$$\mathrm{def}\,(\mathrm{in}\ 1)\,(v0 + v0) = \mathrm{in}\ 1 + \mathrm{in}\ 1$$

$$\mathrm{var}\ vt\,[G\,t] = \mathrm{var}\ vt$$

$$\mathrm{def}\,(v0 + \mathrm{in}\ 1)\,(v0 + v1)\,[\langle \mathrm{in}\ 2\rangle]$$

$$: \mathrm{Tm}\,(\diamond \triangleright \mathrm{Nat})\ \mathrm{Nat}$$

$$: \mathrm{Sub}\ \diamond\ (\diamond \triangleright \mathrm{Nat})$$

$$\text{def}\left(v0 + m1[\langle m2\rangle]\right)$$

$$\left(v0 + v1[\langle m2\rangle^{+}]\right) \overset{+[]}{=}$$

$$\text{def}\left(v0[\langle m2\rangle] + m1[\langle m2\rangle]\right)$$

$$\left(v0[\langle m2\rangle^{+}] + v1[\langle m2\rangle^{+}]\right) \overset{\text{mom}[]}{=}$$

$$\text{def}\left(m2 + m1\right)$$

$$\left(v0 + m2[p]\right) \overset{+(\beta, \text{mod})}{=}$$

$$\text{def}\left(m3\right)\left(v0 + m2\right) \overset{\text{def}\beta}{=}$$

$$v0 + m2[\langle m3\rangle] = m3 + m2 = m5$$

I. Mi az, hogy prog. nyelv? Reszer, Def

II. Típusok: $\Rightarrow$, $\times$, $+$, ind, koind, gen., poli

def : $\mathrm{Tm}\ A \to (\mathrm{Tm}\ A \to \mathrm{Tm}\ B) \to \mathrm{Tm}\ B$

def : $\mathrm{Tm}\ \Gamma\ A \to \mathrm{Tm}\ (\Gamma \triangleright A)\ B \to \mathrm{Tm}\ \Gamma\ B$

környezetek, változók, Γ, vz, vs, $_[_]$

Def. standard modell

$\mathrm{Ty} := \mathrm{Set}$

$\mathrm{Con} := \mathrm{Set}$

$\diamond := T$

$\Gamma \triangleright A := \Gamma \times A$

$\mathrm{Var}\ \Gamma\ A := \Gamma \to A$

$\mathrm{Tm}\ \Gamma\ A := \Gamma \to A$

$\mathrm{vz} := \mathrm{snd}\ (\pi_2)$

$\mathrm{vs}\ t := t \circ \mathrm{fst}$: $\mathrm{Var}\ (\Gamma \triangleright A)\ A = \Gamma \times A \to A$

$\mathrm{vs} := _ \circ \mathrm{fst}$

$\mathrm{true} := \lambda_.\ \mathrm{tt}$

$\mathrm{false} := \lambda_.\ \mathrm{ff}$

$\mathrm{iszero}\ t := \lambda\gamma.\ \mathrm{case}\ t\gamma\ \mathrm{of}$: $\Gamma \to \mathbb{N}$

$0 \to \mathrm{tt}$

$_ \to \mathrm{ff}$

$\mathrm{ite}\ t\ u\ v\ \gamma := \mathrm{ite}\ (t\gamma)\ (u\gamma)\ (v\gamma)$: $\underbrace{t\gamma}_{} : \mathbb{N}$

$\mathrm{Tm}\ (\diamond \triangleright \mathrm{Nat} \triangleright \mathrm{Nat} \triangleright \mathrm{Bool})\ \mathrm{Nat}$

$\parallel$

$T \times \mathbb{N} \times \mathbb{N} \times \mathbb{Z} \to \mathbb{N}$

$$\begin{tikzcd} \Gamma \arrow[r, "t"] & A \\ \Gamma \times B \arrow[u, "\mathrm{fst}"] \arrow[r, "\mathrm{vs}\ t"] & A \arrow[u, equal] \end{tikzcd}$$

$_[_] : \mathrm{Tm}\ \Gamma\ A \to \mathrm{Sub}\ \Delta\ \Gamma \to \mathrm{Tm}\ \Delta\ A$: $(\Gamma \to A) \to (\Delta \to \Gamma) \to (\Delta \to A)$

$\mathrm{Sub}\ \Delta\ \Gamma := \Delta \to \Gamma$

$_[_] := _ \circ _$

$p := \mathrm{fst}$: $\Gamma \times A \to \Gamma$

$\langle t \rangle := \lambda\gamma.\ (\gamma, t\gamma)$: $\Gamma \to \Gamma \times A$ $\quad t : \Gamma \to A$

$\sigma^+ := \lambda(\gamma, a).\ (\sigma\gamma, a)$: $\Delta \times A \to \Gamma \times A$ $\quad \sigma : \Delta \to \Gamma$

$\mathrm{var} := \mathrm{id}$: $(\Gamma \to A) \to (\Gamma \to A)$

$\mathrm{var}\ x\ [p] \overset{?}{=} \mathrm{var}\ (\mathrm{vs}\ x)$

$\parallel \qquad \parallel$

$x \circ \mathrm{fst} = \mathrm{vs}\ x$

$(x \circ \mathrm{fst}) \circ \sigma^+ = \mathrm{var}\ (\mathrm{vs}\ x)\ [\sigma^+] \overset{?}{=}$

$\mathrm{var}\ x\ [\sigma]\ [p]$

$\lambda(\gamma, a).\ x\ (\mathrm{fst}\ (\sigma\gamma, a)) = \lambda(\gamma, a).\ x\ (\sigma\gamma) = x \circ \sigma \circ \mathrm{fst}$

St. $\quad \text{true}\{\Gamma\} \neq \text{false}\{\Gamma\}$

$\text{true}\{\Gamma\} = \text{false}\{\Gamma\} \to \bot$ $\qquad \Gamma = \diamond$

$\downarrow$

$\text{St.}[\![\text{true}]\!] = \text{St.}[\![\text{false}]\!]$

$\updownarrow$

$(\lambda_.tt) = (\lambda_.ff)$

$u \circ (\lambda\gamma.(\gamma, t\,\gamma)) =$

$\parallel$

def $t\,u = u[\langle t \rangle]$

cong : $x = y \to fx = fy$

$(\lambda_.tt) =_{T \to \mathbb{Z}} (\lambda_.ff)$

$f(\lambda_.tt) = tt = ff = f(\lambda_.ff)$

$fx := x$ bin

Normalizálás

jövő óra x perrel más dolog